

Intuitive Understanding of Derivatives

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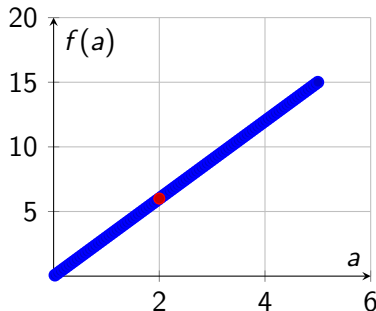
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Why Derivatives Matter in Deep Learning

- Derivative $\equiv$ **slope** $\equiv$ rate of change of a function.
- Powers optimization (gradient descent) and backpropagation.
- We focus on **intuition**, not formal calculus.

A Straight Line: $f(a) = 3a$

- At $a = 2$: $f(a) = 6$.
- Nudge a to 2.001 $\Rightarrow$
 $f(a) = 6.003$.
- Rise = 0.003, run = 0.001.



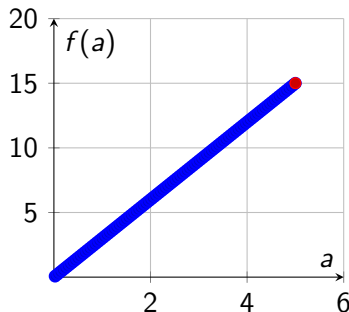
Derivative as Slope

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{0.003}{0.001} = 3$$

- Notation: $\frac{df(a)}{da} = 3$.
- Meaning: small change Δa causes change $\Delta f \approx 3 \Delta a$.

Constant Slope Everywhere

- At $a = 5$: $f(a) = 15$; nudge to $5.001 \Rightarrow 15.003$.
- Straight lines have **constant** slope.
- For $f(a) = 3a$, slope is always 3.



Two Equivalent Notations

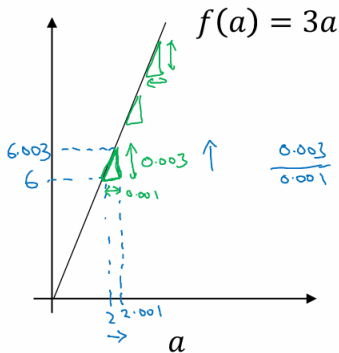
$$\frac{df(a)}{da} = 3 \quad \text{and} \quad \frac{d}{da}f(a) = 3$$

- Both denote instantaneous rate of change of f with respect to a .
- For unit increase in a , $f(a)$ increases by 3.

Infinitesimal Definition (Intuition)

- Formally uses an **infinitesimal** change in a .
- Captures the instantaneous (local) slope.
- For intuition, finite nudges like 0.001 suffice.

Intuition about derivatives



$\rightarrow a = 2$ $f(a) = 6$
 $a = 2.001$ $f(a) = 6.003$

$\rightarrow$ slope (derivative) of $f(a)$
 at $a = 2$ is 3

$\rightarrow a = 5$ $f(a) = 15$
 $a = 5.001$ $f(a) = 15.003$
 slope at $a = 5$ is also 3

$\frac{df(a)}{da} = 3 = \frac{d}{da} f(a)$

$0.001 \leftarrow$
 0.00000001
 0.000000001

Andrew Ng

Summary

- Derivative = slope = rate of change.
- For $f(a) = 3a$, derivative is 3 everywhere.
- This intuition underlies gradient descent.
- Next: functions with **varying** slopes.